Midterm 2

Graded

Student

Daohn Kim

Total Points

98 / 100 pts

Question 1

Question 1 — **28** / 30 pts

1.1 a — 8 / 10 pts

- **− 0 pts** Correct
- ✔ **− 2 pts** first one is wrong
- **− 2 pts** second is wrong
- **− 2 pts** third is wrong
- **− 2 pts** fourth is wrong
- **− 2 pts** fifth is wrong

1.2 b — **10** / 10 pts

- ✔ **− 0 pts** Correct
- **− 3 pts** part a missing
- **− 2 pts** part b missing or wrong
- **− 3 pts** part c missing
- **− 2 pts** part d missing
- **− 2 pts** calculation error

1.3 c — **10** / 10 pts

- ✔ **− 0 pts** Correct
- **− 10 pts** wrong or not done
- **− 5 pts** conduction is missing
- **− 5 pts** conduction equation is wrong

Question 2

Question 2 **35** / 35 pts

2.1 a,b,c **6** / 6 pts

✔ **− 0 pts** Correct

− 1 pt z=1 because it is ideal gas

− 1 pt closed system is missing

− 1 pt energy balance is wrong

− 6 pts not done

2.2 d **4** / 4 pts

✔ **− 0 pts** Correct

− 1 pt v1 = 0.143 m3/kg

− 1 pt v2= 0.858 m3/kg

− 1 pt should be not linear

− 1 pt P2 = 87.35kPa= 0.87 bar

− 1 pt unit error

2.3 e **10** / 10 pts

✔ **− 0 pts** Correct

− 8 pts work equation is wrong

− 2 pts calculation error

− 2 pts P2 is wrong

− 10 pts not done

2.4 f **15** / 15 pts

✔ **− 0 pts** Correct

− 5 pts part a is wrong

− 5 pts part b is wrong

− 5 pts part c is wrong

− 15 pts not done or wrong

− 3 pts deltaT is wrong, it is not zero

− 2 pts Cv for part c is wrong . R should be 8.314/28.9

− 1 pt calculation error

Question 3

Question 3 **35** / 35 pts

3.1 a,b,c **7** / 7 pts

✔ **− 0 pts** Correct

− 1 pt control volume

− 3 pts Energy balance is wrong

− 1 pt energy balance work cant be cancelled

3.2 d **8** / 8 pts

✔ **− 0 pts** Correct

− 2 pts v=0.651m3/kg

− 1 pt calculation error

− 8 pts not done or wrong

− 2 pts not ideal gas

3.3 e **10** / 10 pts

✔ **− 0 pts** Correct

− 10 pts not done or wrong

− 1 pt h1=3022

− 1 pt h2=2782

− 1 pt Q should be negative

− 1 pt Q unit conversion

− 3 pts mdot times (h1-h2)

− 1 pt unit coversion

− 1 pt calculation error

3.4 f **5** / 5 pts

✔ **− 0 pts** Correct

− 1 pt v2=0.651

− 1 pt v1= 0.1254

− 5 pts not done or wrong

− 2 pts v are not marked

3.5 g 5 / 5 pts

✔ − 0 pts Correct

− 5 pts not done

CHE 201 MidTerm Exam 2 October 27th, 2025

NAME: Daohn Kim

Please fill in answers in the space supplied and show all your work to receive full credit.

1. **a) (10 pts)** State whether the following statements are **True of False**

T If a system undergoes a polytropic process isothermally, n=1 for all cases.
T Internal energy is a state property.
T Heat transfer by convection has the unit of joules. Q
F A nozzle is a flow passage of varying cross-sectional area in which the velocity and the pressure of a gas or liquid increases in the direction of flow.
T Enthalpy for the inlet of the turbine is greater than the outlet of the turbine.

$V \leftrightarrow P$ $h_i > h_e$

$-W_{cv} + \dot{m} h_i - \dot{m} h_e = 0$ $\dot{m}(h_i - h_e) = W_{cv} > 0$

b) (10 pts) Determine the volume, in m^3 occupied by 7kg of water at 20MPa and 420°C, using
   a. data from compressibility chart
   b. data from steam table
   c. assuming ideal gas
   d. calculate the percent error

$20 MPa \cdot \frac{10 bar}{1 MPa} = 200 bar$

$420°c + 273 = 693 k$

a. $P_R = \frac{200 bar}{220.9 bar} = 0.9$ $T_R = \frac{693k}{647.3k} = 1.07$ (at $P_R = 0.9$ $T_R = 1.05 \rightarrow Z = 0.6$, $T_R = 1.10 \rightarrow Z = 0.7$)

$Z = 0.65$

$pv = ZRT$ $V = \frac{mZRT}{P} = \frac{(7kg)(0.65)(8.314 kJ/kmol \cdot K)(693K)}{(200bar)(18.02 kg/kmol)(\frac{10^2 kJ}{1 bar\, m^3})} = \boxed{0.0727 m^3}$

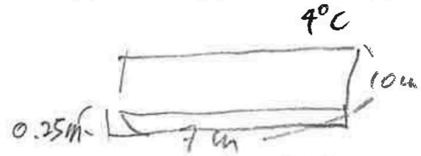
b. at 200bar : $T(°C)$: 400 420 440
$v(m^3/kg)$: 0.00994 0.01222

$\frac{0.01222 - 0.00994}{440 - 400} = \frac{v - 0.00994}{420 - 400}$

$v = 0.01108 m^3/kg$, $V = (7kg)(0.01108 m^3/kg) = \boxed{0.07756 m^3}$

c. $pv = RT$ $V = \frac{mRT}{P} = \frac{(7kg)(8.314 kJ/kmol \cdot K)(693K)}{(200bar)(18.02 kg/kmol)(\frac{10^2 kJ}{1 bar\, m^3})} = \boxed{0.1119 m^3}$

c) (10 points) (FE Exam) The roof of an electrically heated house is 7 m long, 10 m wide, and 0.25 m thick. It is made of a flat layer of concrete whose thermal conductivity is 0.92 W/m.C. During a certain winter night, the temperatures of the inner and outer surfaces of the roof are measured to be 15°C and 4°C, respectively. The average rate of heat loss through the roof that night was
(a) 41 W (b) 177 W (c) 4894 W (d) 5567 W (e) 2834 W



$A = 7m \times 10m = 70m^2$

conduction: $\dot{Q} = -k \cdot A \frac{dT}{dx}$

$= -(0.92 W/m \cdot C)(70m^2) \frac{(15°C - 4°C)}{0.25m}$

$= -2834 W$

$\dot{Q}_{loss} = -\dot{Q} = 2834 W$

d. percent error
compressibility :
$\frac{|0.07756 - 0.0727|}{0.07756} \times 100$
$= 6.27\%$
ideal gas :
$\frac{|0.07756 - 0.1119|}{0.07756} \times 100$
$= 44.3\%$

2. (35 pts)
A piston cylinder device initially contains 1.5m^3 of air at 750kPa and 100°C. The air is now expanded to 9m^3, during which the pressure-volume relationship is $PV^{1.2}$ = constant. The air is modeled as an ideal gas and kinetic and potential energy effects are negligible.

a. (2 pts) Draw a schematic of the system and label the system boundary.
b. (2 pts) State the engineering model
c. (2 pts) Write the simplified energy balance and compressibility factor (Z).
d. (4pts) Draw P-v diagram (Pressure-specific volume)
e. (10 pts) Determine the work (kJ)
f. (15 pts) Determine the heat (kJ) (i) by using Table, (ii) by using constant specific heat values, (iii) by using k= 1.4

Temp. K	c_p (Air)	c_v (Air)	k (Air)
250	1.003	0.716	1.401
300	1.005	0.718	1.400
350	1.008	0.721	1.398
400	1.013	0.726	1.395
450	1.020	0.733	1.391
500	1.029	0.742	1.387
550	1.040	0.753	1.381
600	1.051	0.764	1.376
650	1.063	0.776	1.370
700	1.075	0.788	1.364
750	1.087	0.800	1.359

a)

state 1
1.5m^3
750kPa
100°C

state 2
9m3

$PV^{1.2}$=constant

b) engineering model
→ closed system
→ Ideal gas.
→ $\Delta KE = 0$, $\Delta PE = 0$.

c) E.B
$\Delta E = Q - W$
$\Delta E = \cancel{\Delta KE} + \cancel{\Delta PE} + \Delta U = 0$
$\Delta U = Q - W$

$750kPa \cdot \frac{10^3 Pa}{1 kPa} \cdot \frac{1 bar}{10^5 Pa} = 7.5 bar$

$100°C + 273 = 373K$

Z: state 1) $P_R = \frac{7.5 \text{ bar}}{37.7 \text{ bar}} = 0.2$ $T_R = \frac{373K}{133K} = 2.8 \Rightarrow Z \approx 1.0$

state 2)
$P_2 V_2^{1.2} = P_1 V_1^{1.2}$ $P_2 = \frac{P_1 V_1^{1.2}}{V_2^{1.2}} = \frac{(7.5bar)(1.5m^3)^{1.2}}{(9m^3)^{1.2}} = 0.87 \text{ bar}$

$PV = mRT$: $P_1 V_1 = mRT_1$

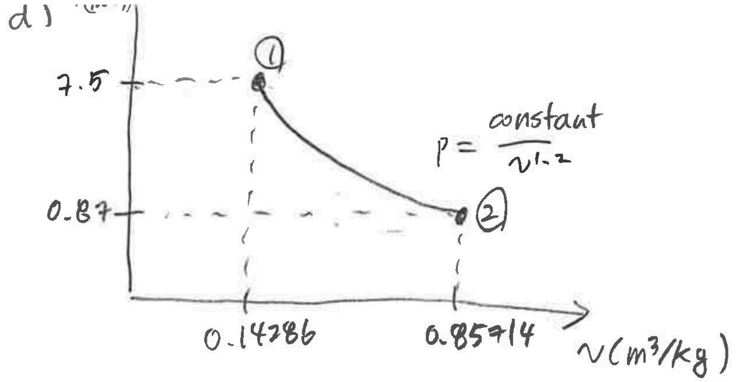
$m = \frac{P_1 V_1}{R T_1} = \frac{(7.5bar)(1.5m^3)\left(\frac{10^2 kJ}{1 bar \cdot m^3}\right)(28.97kg/kmol)}{(8.314 kJ/kmol \cdot K)(373K)}$

$= 10.5 kg$

$T_2 = \frac{P_2 V_2}{mR} = \frac{(0.87bar)(9m^3)\left(\frac{10^2 kJ}{1 bar \cdot m^3}\right)(28.97kg/kmol)}{(10.5kg)(8.314 kJ/kmol \cdot K)} = 259.8K$

$P_R = \frac{0.87bar}{37.7bar} = 0.02$ $T_R = \frac{259.8K}{133K} = 1.95 \approx 2.0 \Rightarrow Z \approx 1.0$

d)



$V_1 = 1.5\,m^3$

$v_1 = 1.5\,m^3 / 10.5\,kg = 0.14286\,m^3/kg$

$V_2 = 9\,m^3$

$v_2 = 9\,m^3 / 10.5\,kg = 0.85714\,m^3/kg$

e) $P_1 \cdot V_1^{1.2} = \text{constant}$ $\qquad V^{-1.2+1}_{\;-0.2}$

$= (7.5\,bar)(1.5\,m^3)^{1.2} = 12.2$

$W = \int p\,dV = \int_{1.5}^{9} \frac{12.2}{V^{1.2}}\,dV = 12.2\left[-\frac{1}{0.2}V^{-0.2}\right]_{1.5}^{9}$

$= \frac{-12.2}{0.2}\left[9^{-0.2} - 1.5^{-0.2}\right] = 16.94\,bar\cdot m^3 \cdot \left(\frac{10^2\,kJ}{1\,bar\cdot m^3}\right)$

$= \boxed{1694\,kJ}$

f) i) by the table

373k → 370k $\quad$ 380k

$\qquad$ 264.46 $\quad$ 271.69 (kJ/kg)

$\frac{271.69 - 264.46}{380 - 370} = \frac{u - 264.46}{373 - 370}$

$u = 266.63\,kJ/kg$

259.8k ≈ 260k ⇒ 185.45 kJ/kg

$\Delta U = \Delta u \cdot m = (185.45\,kJ/kg - 266.63\,kJ/kg)(10.5\,kg) = -852\,kJ$

$\Delta U = Q - W \quad Q = W + \Delta U = 1694\,kJ - 852\,kJ = \boxed{842\,kJ}$

ii) $\Delta U = m \cdot C_v \cdot \Delta T$

$= (10.5\,kg)(1.005)(373\,k - 260\,k)\frac{1}{28.97\,kg/kmol} = -41.1\,kJ$

$Q = 1694\,kJ - 41.1\,kJ = \underline{1652.9\,kJ}$

iii) $\Delta U = m \cdot \frac{R}{k-1} \cdot \Delta T = (10.5\,kg)\frac{(8.314\,kJ/kmol\cdot k)(260 - 373)\,k}{(1.4-1)(28.97\,kg/kmol)} = -851\,kJ$

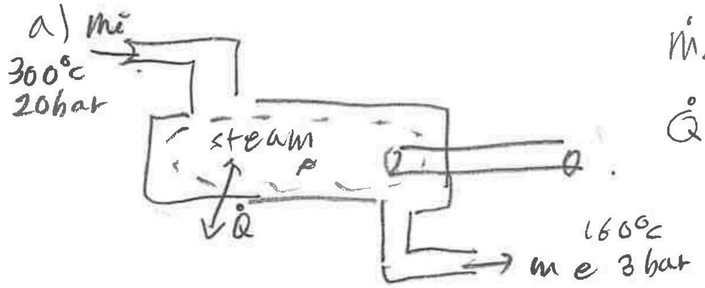
$Q = 1694\,kJ - 851\,kJ = \boxed{846\,kJ}$

3. (35 points)

A steam turbine operating at steady state is providing power. Steam enters turbine at 300^0C and 20 bar. The steam exits the turbine at 160^0C and 3 bar. The mass flow rate of the steam at the turbine inlet is 5.5 kg/s. The rate of heat transfer from the turbine to its surroundings is 80 kJ/min. Kinetic and potential energy changes are negligible.

a) (2 pts) Draw a schematic of the system and label the system boundary.
b) (2 pts) State the engineering model
c) (3 pts) Write the simplified energy rate balance
d) (8 pts) Determine the volumetric flowrate of the steam at the turbine exit, in m3/s.
e) (10 pts) Determine the power in kW.
f) (5 pts) Sketch the process from State 1 to State 2 on a P-v diagram.
g) (5 pts) Explain how you would increase the power output.

a)



$\dot{m}_i = 5.5 \text{ kg/s}$

$\dot{Q} = -80 \text{ kJ/min} \cdot \left(\frac{1 \text{ min}}{60 \text{ s}}\right) = -1.33 \text{ kJ/s}$

b) engineering model
→ control volume
→ one inlet one outlet
→ $\Delta KE = 0$ $\Delta PE = 0$
→ steady state

M·B

$\frac{dm_{cv}}{dt} = \sum \dot{m}_i - \sum \dot{m}_e$

$\dot{m}_i = \dot{m}_e$

c) E·B) $\cancel{\frac{dE_{cv}}{dt}}^{0} = \dot{Q} - \dot{W}_{cv} + \dot{m}_i\left(h_i + \cancel{\frac{V_i^2}{2}} + \cancel{gz_i}\right) - \dot{m}_e\left(h_e + \cancel{\frac{V_e^2}{2}} + \cancel{gz_e}\right)$

$\dot{Q} - \dot{W}_{cv} + \dot{m}_i h_i - \dot{m}_e h_e = 0$

$\dot{Q} - \dot{W}_{cv} + \dot{m}_i(h_i - h_e) = 0$ (by M-B)

d) $\dot{m}_i = \frac{\dot{V}_i}{v_i} = \frac{\dot{V}_e}{v_e}$ $\dot{V}_e = \dot{m}_i v_e$

at 160°C, 3 bar, $v_e = 0.651$ m³/kg

$\dot{V}_e = (5.5 \text{ kg/s}) \cdot (0.651 \text{ m}^3/\text{kg}) = \boxed{3.58 \text{ m}^3/\text{s}}$

e) $\dot{W}_{cv} = \dot{Q} + \dot{m}_i(h_i - h_e)$

h_i) T : 280°C · 300°C   320°C
h : 2976.4 h_i 3069.5 (kJ/kg)

$\frac{3069.5 - 2976.4}{320 - 280} = \frac{h_i - 2976.4}{\cancel{320}300 - 300}$

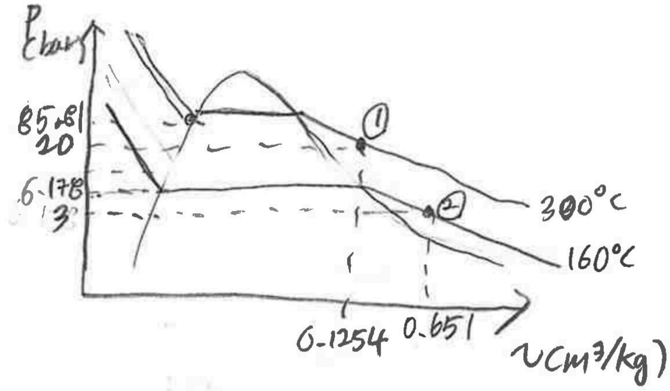
$h_i = 3022.95$ kJ/kg

$h_e = 2782.3$ kJ/kg

$= -1.33\ kJ/s + (5.5\ kg/s)(3022.45\ kJ/kg - 2782.3\ kJ/kg)$

$= 1322.2\ kJ/s$

$= \boxed{1322.2\ kW}$

f) 

v_{r} : 280°C 320°C
0.1200 0.1308 (m³/kg)

$v_i = 0.1254\ m^3/kg$

g) To increase the power output, the heat transfer should reduce to its surroundings. If the heat loss decreases, the power output will increase.